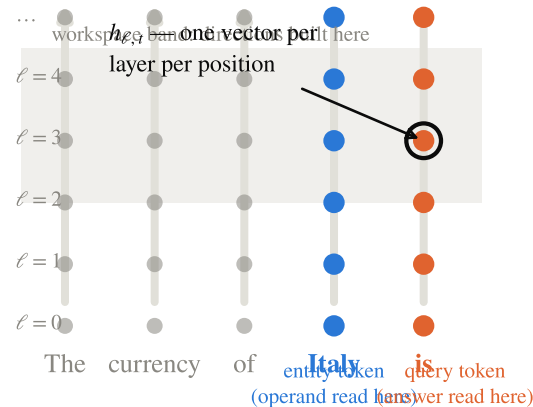


The state is a grid: layers $\times$ positions



One object, applied three ways

at each band layer ℓ , average the query-token states:

$$v_{\ell}(\text{op}) = \text{mean}_o [h_{\ell, -1}(o, \text{op}) - \text{mean}_k h_{\ell, -1}(o, k)]$$

$\equiv$ the ANOVA operator main effect $b_{\ell}(\text{op})$ (verified: dev. 10^{-8})

steer add $\alpha (v_{\ell}(B) - v_{\ell}(A))$ at the query token $\rightarrow$ margin flips; generates at calibrated α

compose replace $h_{\ell, -1}$ with $\mu_{\ell} + a_{\ell}(o) + b_{\ell}(k) \rightarrow$ generates at the clean ceiling

decode classify / ANOVA the same states $\rightarrow$ factorization, layer profiles

Prompt $\rightarrow$ token $\times$ layer grid of residual states $\rightarrow$ per-layer operator direction by averaging at the query token $\rightarrow$ applied by addition (steering), by replacement (composition), or read out (ANOVA).